



2026

MATHEMATICS

Specimen Paper 3 of 4

Topics: Surds • Quadratic Equations • Circle Theorems • Mensuration

Time Allowed: 2 hours **Total Marks:** 100 **Calculators:** Permitted

Index Number: _____

Formulae Sheet

Area & Perimeter

Rectangle: $A = l \times w$, $P = 2(l+w)$

Triangle: $A = \frac{1}{2}bh$

Circle: $A = \pi r^2$, $C = 2\pi r$

Trapezium: $A = \frac{1}{2}(a+b)h$

Volume

Cuboid: $V = lwh$

Cylinder: $V = \pi r^2 h$

Cone: $V = \frac{1}{3}\pi r^2 h$

Sphere: $V = \frac{4}{3}\pi r^3$

Trigonometry

$\sin \theta = \text{opp/hyp}$ $\cos \theta = \text{adj/hyp}$

$\tan \theta = \text{opp/adj}$

Other

Pythagoras: $a^2 + b^2 = c^2$

Quadratic: $x = \frac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$

Standard Form: $a \times 10^n$, $1 \leq a < 10$

Circle Theorems

- Angle at centre = $2 \times$ angle at circumference (same arc)
- Angle in a semicircle = 90°
- Opposite angles of a cyclic quadrilateral sum to 180°
- Tangent to a circle is perpendicular to the radius at the point of contact
- Angles in the same segment are equal

Instructions to Candidates

1. Write your index number in the space provided above.
2. Answer **ALL** questions in Sections A, B and C.
3. For Section A, tick **ONE** box only per question.
4. For Sections B and C, show all working clearly.
5. Marks are shown in brackets [] at the right of each question or part.
6. You may use a calculator for this paper.

7. This paper contains 28 questions. **Total: 100 marks.**

SECTION A — Multiple Choice [10 marks] Tick ONE answer for each question. 1 mark each.

- 1 Which is the simplified form of $\sqrt{50}$? [1]
- ☐ A $5\sqrt{2}$
- ☐ B $25\sqrt{2}$
- ☐ C $10\sqrt{5}$
- ☐ D $2\sqrt{25}$
- 2 The value of $(\sqrt{5})^2$ is: [1]
- ☐ A 5
- ☐ B $\sqrt{10}$
- ☐ C 25
- ☐ D 10
- 3 The roots of $x^2 - 5x + 6 = 0$ are: [1]
- ☐ A $x = 2$ and $x = 3$
- ☐ B $x = -2$ and $x = -3$
- ☐ C $x = 1$ and $x = 6$
- ☐ D $x = -1$ and $x = -6$
- 4 The angle subtended by an arc at the centre of a circle is *related* to the angle subtended by the same arc at the circumference as: [1]
- ☐ A Angle at centre = $2 \times$ angle at circumference
- ☐ B Angle at centre = angle at circumference
- ☐ C Angle at centre = $\frac{1}{2} \times$ angle at circumference
- ☐ D Angle at centre = $3 \times$ angle at circumference
- 5 In a cyclic quadrilateral PQRS, the sum of opposite angles is: [1]
- ☐ A 90°
- ☐ B 180°
- ☐ C 270°
- ☐ D 360°
- 6 The area of a circle with radius 7 cm is: [1]
- ☐ A $49\pi \text{ cm}^2$
- ☐ B $14\pi \text{ cm}^2$

- ☐ C $7\pi \text{ cm}^2$
☐ D $196\pi \text{ cm}^2$

7 The simplified form of $\sqrt{12}$ is:

[1]

- ☐ A $2\sqrt{3}$
☐ B $3\sqrt{4}$
☐ C $4\sqrt{3}$
☐ D $6\sqrt{2}$

8 The discriminant ($b^2 - 4ac$) of the equation $x^2 + 2x + 1 = 0$ is:

[1]

- ☐ A 8
☐ B 4
☐ C 0
☐ D -4

9 A tangent to a circle meets the radius drawn to the point of contact at an angle of:

[1]

- ☐ A 45°
☐ B 60°
☐ C 90°
☐ D 180°

10 The volume of a cylinder with radius 3 cm and height 5 cm is:

[1]

- ☐ A $15\pi \text{ cm}^3$
☐ B $30\pi \text{ cm}^3$
☐ C $45\pi \text{ cm}^3$
☐ D $90\pi \text{ cm}^3$

SECTION B — Short Structured Questions [48 marks] Show all working.

11 Simplifying Surds

[3]

(a) Simplify $\sqrt{72}$.

[1]

(b) Simplify $\sqrt{18} + \sqrt{8}$.

[2]

12 Quadratic Equations — Factorisation

[3]

(a) Solve $x^2 + x - 6 = 0$ by factorisation.

[2]

$x =$ or $x =$

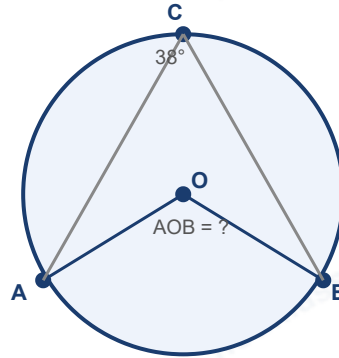
(b) State the y-intercept of the curve $y = x^2 + x - 6$.

[1]

13 Circle Theorems — Angle at Centre

[4]

In the diagram, O is the centre of the circle. A, B and C are points on the circumference. Angle $ACB = 38^\circ$.



- (a) State the circle theorem that relates angle AOB to angle ACB.

[1]

- (b) Calculate angle AOB.

[1]

Angle AOB = $^\circ$

- (c) Triangle AOB is isosceles. Calculate angle OAB.

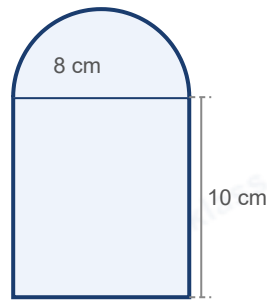
[2]

Angle OAB = $^\circ$

14 Mensuration — Composite Shape

[4]

The shape below consists of a rectangle with a semicircle on top. The rectangle has width 8 cm and height 10 cm. The semicircle has diameter 8 cm.



- (a) Calculate the area of the rectangle.

[1]

Area of rectangle = cm^2

- (b) Calculate the area of the semicircle. Give your answer in terms of π .

[1]

Area of semicircle = cm^2

- (c) Calculate the total area of the composite shape. Give your answer correct to 1 decimal place.

[2]

Total area = cm^2

15 Quadratic Formula

[4]

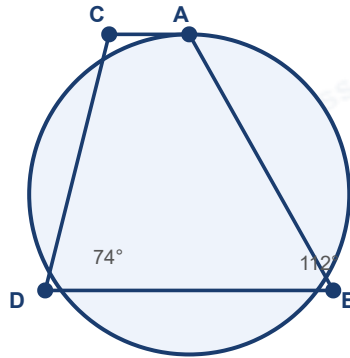
Solve $x^2 + 5x + 3 = 0$ using the quadratic formula. Give your answers to 2 decimal places.

$x =$ or $x =$

16 Circle Theorems — Cyclic Quadrilateral

[4]

ABCD is a cyclic quadrilateral. Angle $ABC = 112^\circ$ and angle $BCD = 74^\circ$.



(a) Calculate angle ADC. State the circle theorem you use.

[2]

Angle ADC = $^\circ$

(b) Calculate angle DAB.

[2]

Angle DAB = °

17 Volume of Solids

[4]

- (a) A cylinder has radius 5 cm and height 12 cm. Calculate its volume in terms of π .

[2]

Volume = cm^3

- (b) A cone has the same base radius (5 cm) and same height (12 cm) as the cylinder in (a). What fraction of the cylinder's volume is the cone's volume?

[2]

18 Surds — Expanding and Rationalising

[5]

- (a) Expand and simplify $(2 + \sqrt{3})(4 - \sqrt{3})$.

[3]

- (b) Rationalise the denominator of $\frac{6}{\sqrt{3}}$.

[2]

19 Quadratic Word Problem

[4]

A rectangular garden has length x cm and width $(x - 3)$ cm. The area of the garden is 40 cm^2 .

- (a) Form a quadratic equation in x . Show that it simplifies to $x^2 - 3x - 40 = 0$.

[2]

- (b) Solve $x^2 - 3x - 40 = 0$ by factorisation. Hence find the dimensions of the garden.

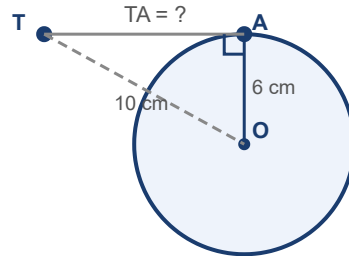
[2]

Length = cm Width = cm

20 Circle Theorem — Tangent

[4]

TA is a tangent to the circle at A. O is the centre. OA = 6 cm and OT = 10 cm.



- (a) Explain why angle OAT = 90° .

[1]

- (b) Calculate the length TA.

[3]

TA = cm

21 Mensuration — Cone and Cylinder

[5]

A solid consists of a cylinder of radius 4 cm and height 9 cm, with a cone of the same base radius and height 6 cm placed on top.

- (a) Calculate the volume of the cylinder. Leave your answer in terms of π .

[1]

- (b) Calculate the volume of the cone. Leave your answer in terms of π .

[2]

- (c) Calculate the total volume of the solid. Give your answer to 3 significant figures.

[2]

Total volume = cm^3

22 Surds — Arithmetic

[4]

- (a) Simplify $3\sqrt{5} \times 2\sqrt{5}$.

[1]

- (b) Simplify $(1 + \sqrt{2})^2$.

[3]

SECTION C — Long Structured Questions [42 marks] Show all working clearly.

23 Surds — Extended

[7]

(a) Simplify $\sqrt{75} - \sqrt{27} + \sqrt{3}$.

[3]

(b) Expand and simplify $(\sqrt{6} + \sqrt{2})(\sqrt{6} - \sqrt{2})$.

[2]

(c) Given that $\sqrt{k} = 3\sqrt{5}$, find the value of k .

[2]

$k =$

24 Quadratic Equations — Extended

[7]

- (a) Solve $2x^2 + 5x - 3 = 0$ by factorisation.

[3]

$x =$ or $x =$

- (b) The equation $x^2 + 6x + k = 0$ has exactly one solution (a repeated root).

(i) Find the value of k .

[2]

$k =$

(ii) Hence find the solution.

[1]

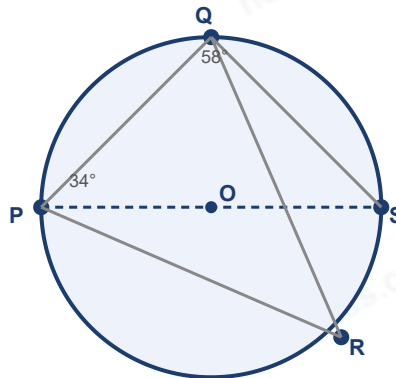
$x =$

- (c) The curve $y = x^2 - 4x + 3$ crosses the x -axis. State the x -coordinates of these crossing points.

[1]

25 Circle Theorems — Multi-step**[7]**

O is the centre of the circle. Points P, Q, R and S lie on the circumference. Angle QPR = 34° , angle PQS = 58° , and PS is a diameter.



- (a) Find angle QSP. Give a reason.

[2]

Angle QSP = $^\circ$

- (b) Find angle PSR. Give a reason.

[2]

Angle PSR = $^\circ$

- (c) Find angle QPS. Show full reasoning.

[3]

Angle QPS = °

26 Mensuration — Composite 3D Shape

[7]

A storage tank consists of a cylinder of radius 2 m and height 5 m, with a hemisphere (half-sphere) on top of the same radius.

- (a) Write down the formula for the volume of a hemisphere in terms of r .

[1]

- (b) Calculate the volume of the cylindrical part. Leave your answer in terms of π .

[2]

- (c) Calculate the volume of the hemispherical top. Leave your answer in terms of π .

[2]

- (d) Calculate the total volume of the tank. Give your answer to 3 significant figures.

[2]

Total volume = m^3

27 Quadratic Application

[7]

A ball is thrown upward from the top of a cliff. Its height h metres above sea level at time t seconds is given by $h = -5t^2 + 20t + 60$.

- (a) Find the initial height of the ball (when $t = 0$).

[1]

Initial height = m

- (b) Find the height of the ball when $t = 2$ seconds.

[2]

Height at $t=2$ = m

- (c) The ball hits the sea when $h = 0$. Form the equation $-5t^2 + 20t + 60 = 0$ and simplify to $t^2 - 4t - 12 = 0$. [1]

- (d) Solve $t^2 - 4t - 12 = 0$ by factorisation. Hence find the time at which the ball hits the sea. Explain why one solution is rejected. [3]

Time = s

28 Circle Theorems + Mensuration [7]

A circle has centre O and radius 10 cm. Chord AB has length 16 cm.

- (a) Draw a perpendicular from O to chord AB, meeting AB at M. Explain why OM bisects AB. [1]

- (b) Calculate the length OM. [3]

OM = cm

- (c) Calculate the area of triangle OAB.

[1]

Area of OAB = cm²

- (d) Angle AOB = $2 \arcsin(8/10)$. Calculate angle AOB in degrees.

[1]

Angle AOB = °

- (e) Calculate the area of the minor sector AOB. Give your answer to 1 decimal place.

[1]

Area of sector = cm²

Mark Scheme — Specimen Paper 3

Marking guidance: M = method mark. A = accuracy mark. B = unconditional mark. ft = follow through.

Section A — MCQ Answers

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
A	A	A	A	B	A	A	C	C	C

Q	Answer / Working	Marks	Notes
11(a)	$\sqrt{72} = \sqrt{(36 \times 2)} = 6\sqrt{2}$	1	B1
11(b)	$\sqrt{18} + \sqrt{8} = 3\sqrt{2} + 2\sqrt{2} = 5\sqrt{2}$	2	M1 simplify each; A1
12(a)	$(x+3)(x-2)=0$; $x=-3$ or $x=2$	2	M1 factorisation; A1
12(b)	-6 (set $x=0$)	1	B1
13(a)	Angle at centre = $2 \times$ angle at circumference (same arc)	1	B1
13(b)	$AOB = 2 \times 38^\circ = 76^\circ$	1	B1 ft
13(c)	$OA=OB$ (radii); angle $OAB = (180-76) \div 2 = 52^\circ$	2	M1 isosceles; A1
14(a)	$8 \times 10 = 80 \text{ cm}^2$	1	B1
14(b)	$\frac{1}{2}\pi(4)^2 = 8\pi \text{ cm}^2$	1	B1
14(c)	$80 + 8\pi = 80 + 25.13... \approx 105.1 \text{ cm}^2$	2	M1; A1 (105.1)
15	$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(1)(12)}}{2} = \frac{-(-5) \pm \sqrt{25 - 48}}{2}$; $x \approx -0.70$ or $x \approx -4.30$	4	M1 formula; M1 discriminant; A1 each root
16(a)	$ADC = 180 - 112 = 68^\circ$; opposite angles of cyclic quad sum to 180°	2	M1 theorem; A1
16(b)	$DAB = 180 - 74 = 106^\circ$	2	M1; A1
17(a)	$V = \pi(5)^2(12) = 300\pi \text{ cm}^3$	2	M1; A1
17(b)	Cone $V = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times 300\pi = 100\pi$; fraction $= 100\pi / 300\pi = 1/3$	2	M1; A1
18(a)	$(2 + \sqrt{3})(4 - \sqrt{3}) = 8 - 2\sqrt{3} + 4\sqrt{3} - 3 = 5 + 2\sqrt{3}$	3	M1 expand 4 terms; M1 collect; A1
18(b)	$6/\sqrt{3} \times \sqrt{3}/\sqrt{3} = 6\sqrt{3}/3 = 2\sqrt{3}$	2	M1 multiply by $\sqrt{3}/\sqrt{3}$; A1

Q	Answer / Working	Marks	Notes
19(a)	$x(x-3)=40$; $x^2-3x=40$; $x^2-3x-40=0$	2	M1; A1 showing steps
19(b)	$(x-8)(x+5)=0$; $x=8$ (reject $x=-5$); Length=8cm, Width=5cm	2	M1 factorisation; A1
20(a)	Tangent is perpendicular to radius at point of contact (circle theorem)	1	B1
20(b)	$TA^2=OT^2-OA^2=100-36=64$; $TA=8$ cm	3	M1 Pythagoras; M1 substitution; A1
21(a)	$V=\pi(4)^2(9)=144\pi$ cm ³	1	B1
21(b)	$V=\frac{1}{3}\pi(4)^2(6)=32\pi$ cm ³	2	M1; A1
21(c)	$(144+32)\pi=176\pi\approx 552.9\approx 553$ cm ³	2	M1 addition; A1 3 s.f.
22(a)	$3\sqrt{5}\times 2\sqrt{5}=6\times 5=30$	1	B1
22(b)	$(1+\sqrt{2})^2=1+2\sqrt{2}+2=3+2\sqrt{2}$	3	M1 expand; M1 simplify $\sqrt{2}\times\sqrt{2}$; A1
23(a)	$\sqrt{75}=5\sqrt{3}$; $\sqrt{27}=3\sqrt{3}$; $5\sqrt{3}-3\sqrt{3}+\sqrt{3}=3\sqrt{3}$	3	M1 simplify each; M1 collect; A1
23(b)	$(\sqrt{6})^2-(\sqrt{2})^2=6-2=4$ (difference of squares)	2	M1; A1
23(c)	$\sqrt{k}=3\sqrt{5}$; $k=(3\sqrt{5})^2=9\times 5=45$	2	M1 squaring both sides; A1
24(a)	$(2x-1)(x+3)=0$; $x=\frac{1}{2}$ or $x=-3$	3	M1 trial; M1 factorisation; A1
24(b) (i)	Discriminant=0: $36-4k=0$; $k=9$	2	M1; A1
24(b) (ii)	$x=-b/2a=-6/2=-3$	1	B1 ft
24(c)	$y=x^2-4x+3=(x-1)(x-3)=0$; $x=1$ and $x=3$	1	B1
25(a)	Angle QSP=34°; angles in the same segment are equal (QP is chord, QPR and QSP on same side)	2	M1 theorem; A1
25(b)	Angle PSR=90°; angle in a semicircle (PS is diameter)	2	M1 theorem; A1
25(c)	In triangle PQS: angle QPS=180-58- (angle QSP)=180-58-34=88°; [or use: angle QPS+angle PQS+angle QSP=180]	3	M1 triangle sum; M1 correct angles; A1
26(a)	$V = \frac{1}{2} \times \frac{4}{3} \pi r^3 = \frac{2}{3} \pi r^3$	1	B1
26(b)	$V=\pi(2)^2(5)=20\pi$ m ³	2	M1; A1

Q	Answer / Working	Marks	Notes
26(c)	$V = \frac{2}{3}\pi(2)^3 = \frac{2}{3}\pi \times 8 = \frac{16}{3}\pi \text{ m}^3$	2	M1; A1
26(d)	$(20+16/3)\pi = (76/3)\pi \approx 79.6 \text{ m}^3$	2	M1; A1 3 s.f.
27(a)	$h=60 \text{ m}$ (when $t=0$)	1	B1
27(b)	$h = -5(4)+20(2)+60 = -20+40+60=80 \text{ m}$	2	M1; A1
27(c)	Divide by -5 : $t^2-4t-12=0$	1	B1
27(d)	$(t-6)(t+2)=0$; $t=6$ or $t=-2$; reject $t=-2$ (time cannot be negative); ball hits sea at $t=6 \text{ s}$	3	M1 factorisation; A1 both roots; B1 correct rejection
28(a)	Perpendicular from centre bisects chord (circle theorem)	1	B1
28(b)	$AM=8$; $OM=\sqrt{(10^2-8^2)}=\sqrt{(100-64)}=\sqrt{36}=6 \text{ cm}$	3	M1 $AM=8$; M1 Pythagoras; A1
28(c)	$\text{Area} = \frac{1}{2} \times 16 \times 6 = 48 \text{ cm}^2$	1	B1 ft
28(d)	Angle $AOB=2 \arcsin(8/10)=2 \times 53.13^\circ \approx 106.3^\circ$	1	B1
28(e)	$\text{Area} = (\theta/360) \times \pi r^2 = (106.3/360) \times \pi \times 100 \approx 92.9 \text{ cm}^2$	1	B1 ft (accept 92–93)